

Skin diseases Identification via Deep Learning and CNN: Using Optimized ResNet50 Architecture

Ankur Kumar

Dept. Of Computer Science and Engineering
KIET Group of Institutions
Ghaziabad, India
ankurhal22@gmail.com

Anmol Kumar

Dept. Of Computer Science and Engineering
KIET Group of Institutions
Ghaziabad, India
anmolkumar100bc@gmail.com

Anshul Rai

Dept. of Computer Science and Engineering
KIET Group of Institutions
Ghaziabad, India
raianshul33@gmail.com

Pushpendra Kumar

Dept. of Computer Science and Engineering
KIET Group of Institution
Ghaziabad, India
pushpendra.kumar@kiet.edu

Abhishek Yadav

Dept. Of Computer Science and Engineering
KIET Group of Institutions
Ghaziabad, India
abhii199works@gmail.com

Abstract—Skin disease is one of the most common human ailments but the access to the specialist dermatologist is limited. This paper present an optimized ResNet50 model for multi class skin disease classification. The main goal of this project is to improve the accuracy and generalization through a combination of transfer learning, data augmentation and regularization techniques. Automated image based classification model can provide early diagnosis and lower the risk or damage to the body. We have fine tuned a ResNet50 model, pretrained on ImageNet. The model was trained on a dataset of 22 different skin diseases. We have optimized it by extensive data augmentation techniques like random flip, rotation, zoom, contrast, a multi layer classifier and a learning rate scheduler. Training is further stabilized with early stopping. The proposed model has achieved an overall accuracy of 36.4%, weighted average F1-score of 0.31 and recall 0.27. The result shows a challenge of class imbalance in medical image classification and need for more sophisticated strategies like using class weighted and artificial data generation. Considering the size of the dataset, the optimized RestNet 50 demonstrate good performance. Future work will focus on addressing class imbalance and deploying model in clinical operations.

I. INTRODUCTION

Millions of people get affected by skin diseases every year and across all age group. [1] Conditions like acne, eczema, psoriasis and skin cancer not only affect the life of people but also pose a challenge in diagnosis due to diverse clinical presentation and overlap in visual appearance. And also access to specialist dermatologist is limited in many regions, that leads to incorrect or delayed diagnosis. An automated image based diagnosis model can help in assiting professional and provide preliminary screening. [2]Through Deep learning, par-

ticularly Convolutional Neural Network (CNN) has changed medical image analysis significantly. Models like RedNet, VCG and Inception has demonstrated remarkable performance in skin lesion classification and sometimes surpassing human expectation. Early accurate diagnosis is important, especially for skin cancer like life risking diseases. Therefore we need models like ResNet50 which is very popular because of it's balance between depth and computational efficiency. But we can not directly apply such models in real world clinical problems, there will be some problems to it. First of all medical dataset are often small, imbalanced and contain variability in lightning, scale and other imaging condition. And also the cost of acquiring a labeled dataset for rare disease lead to severe imbalance in classes of these variable disease dataset. This imbalance can make the model bias towards majority of the classes, resulting in poor generalization and unreliable prediction for uncommon diseases. To address such problems, researchers have explored various optimization strategies, including transfer learning with pre-trained weight, data augmentation, architectural modification and regularization technique etc. And also steps involved in the complete processing need to be optimized. Here we present an optimized version of ResNet50 for skin disease classification, improving the transfer learning with robust regularization. First of all fine tuned the selectively unfreeze later layer of ResNet50 pretrained on ImagNet, which helps in adaption to skin specific feature and preserving the generic visual knowledge. [3] We have also worked on regularization pipeline including data augmentation, a multi layer classifier early

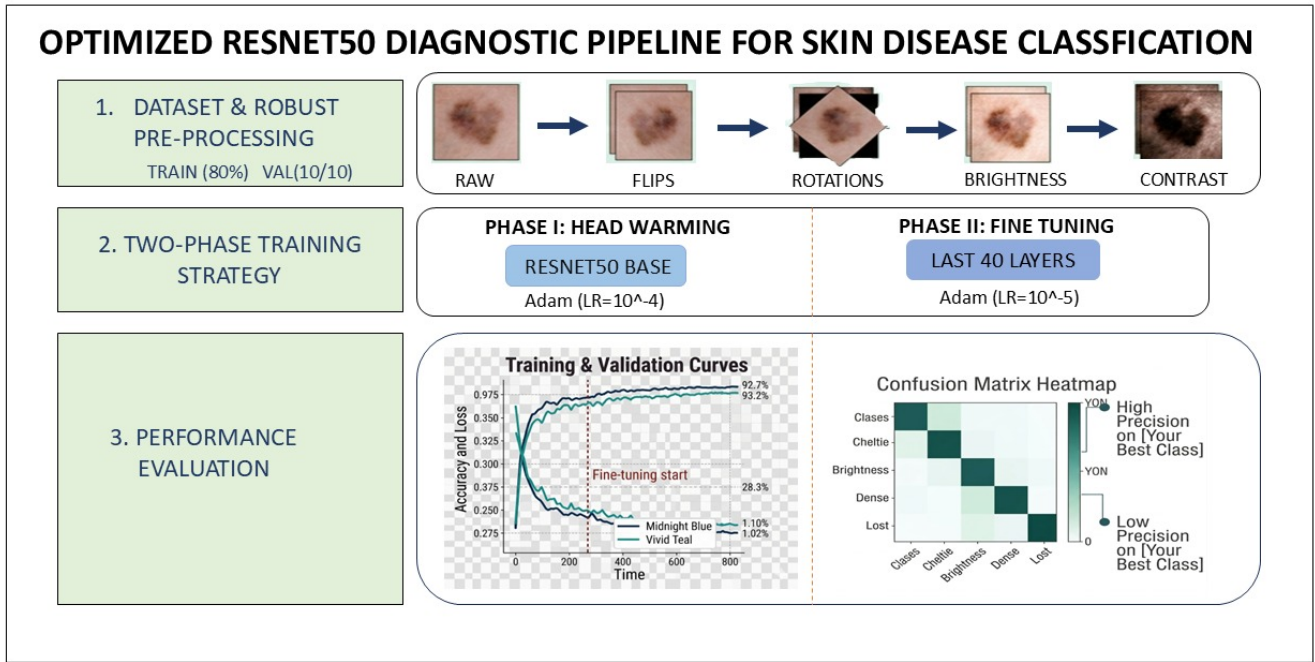


Fig. 1. Pipelined for Model

stopping and learning rate scheduling. We have trained model on publicly available dataset. The result shows our model achieved moderate accuracy, higher than previous version. The only challenge that we have faced majorly here is the class imbalance.

II. LITERATURE REVIEW

One the most popular area in the research for Artificial Intelligence (AI) is Healthcare. From basic machine learning (ML) classifiers to more complex Deep Learning architectures, there is a huge opportunity in this field. And we can clearly see there is a significant progress in how AI is applied to various areas within healthcare system [4]. Most of the initial AI in dermatology based on binary classifications (i.e. distinguishing between healthy skin and lesions). But now it has become common for dermatologists to evaluate skin lesions using a multi-class severity grading system and for AI systems to identify specific lesions [5].

A. Traditional Approaches to Skin Disease Classification

Early computer diagnosis system for skin diseases rely on hand crafted features like color, shape descriptor, texture and dermoscopic image and fed then into classical machine learning classifier such as support vector machine, k- nearest neighbours and random forest. These system demonstrated feasibility but their performance was limited. These systems or models were very basic and cannot do much operations.

B. Deep Learning in Skin disease Detection

The emergence of deep convolutional neural network marked a significant shift in medical image analysis [6]. Researchers have shown that a single CNN could achieve dermatologist level accuracy in classifying skin lesion. But it need huge dataset. ResNet50 became very popular due to skip connections that allow training of very deep network without vanishing gradients. In the recent year, EfficientNet and MobileNet have shown great performance for mobile base platform which are lighter in size and in parameters. Other architectures such as DenseNet, Vision Transformer have also shown promising result. Deep Learning has helped a lot in optimizing the model and improving the accuracy and precision. [7] Also for other living organisms like plant we can use deep learning in monitoring there health [8].

C. ResNet50 and Transfer Learning

Transfer Learning is a technique which uses a pretrained model as a starting point for another related task. It is used in medical imaging because of pre training on large dataset. [9] Instead of training from scratch, here we require less resource and less labeled data. [10] Through transfer learning we start working from an pretrained model which saves lot of time and provide computational efficiency. Through ResNet50, our strategy is to freeze early layer to capture low level features and fine tuned later layer to adapt to the target domain [11]. We can also make a hybrid model which have features of both

model and more models, this will improve the accuracy and efficiency of the model.

D. Optimization technique for CNN based Medical Image diagnosis

There are lot of factors on which the processing depends and fine tuning them will improve the performance of the model. Here for the optimization of ResNet50 we have worked on following fields:

- **Data Augmentation:** Medical datasets are sometimes small, tilted, patchy etc. So we have done some augmentation geometric transformations like: rotation, flipping, zoom, colour adjustments, and more advanced techniques like Cut and mix which increases effective sample size and improves robustness. In skin lesion classification, augmentation is very important to handle different images that vary in lighting, scale, and orientation.
- **Regularisation:** Dropout and batch normalization are standard techniques to reduce overfitting. Dropout method randomly deactivates neurons during training, forcing the network to learn redundant representations. This method help in learning new features and stop memorizing general things.
- **Learning Rate Scheduling:** Adaptive learning rate techniques help the model to converge to a better option by reducing the learning rate in case of validation loss plateaus. This method is often combined with early stopping to prevent overfitting.

III. METHODOLOGY

We have developed a structured pipeline for the proposed model focusing on increasing the accuracy and minimizing required computational resources. The overall pipeline is divided into following steps:

A. Dataset Description

We have used a publicly available skin disease dataset from KaggleHub (creatorai/skin-disease). The dataset contains dermoscopic and clinical images of various skin condition, grouped into 22 distinct disease categories [12]. For our model we have divided dataset into 80/10/10 where, 80% of the images are used for training, 10% for validation and 10% for testing. All the images are resized into 224 X 224 pixel to match the input size of ResNet50. The dataset have some class imbalance which reflect the real life condition where certain diseases are much more common than other diseases.

B. Pre-processing and Data Augmentation

All the images are pre-processed using some specific functions of ResNet50 which scales pixel values to the range expected by the pre-trained network. To increase the diversity of the training dataset and reduce overfitting, we have applied some data augmentation technique on training set such as: Random horizontal and vertical flips, Random rotation upto 20 degree, Random zoom upto 20% and Random contrast adjustment (around 10%). These augmentation techniques

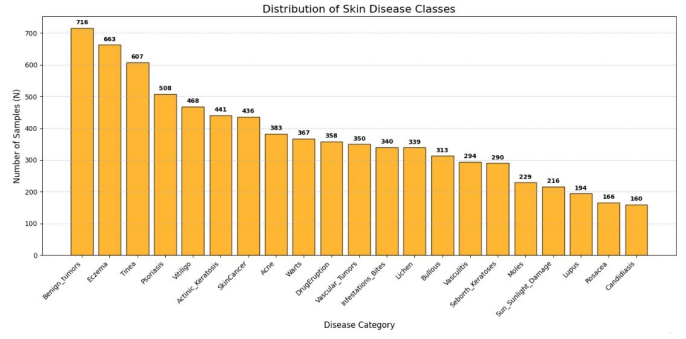


Fig. 2. Dataset Distribution

were applied by TensorFlow's Sequential augmentation layer. The validation and test sets receive only the preprocessed input.

- **Normalization Layer:** It is an input preprocessing step that make ensure that the images are scaled in the same way as during the original ImageNet training.
- **Global Average Pooling:** Global Average Pooling (GAP) in ResNet50 is widely used for skin disease detection because it reduces overfitting, improves generalization, and makes the model more robust to variations in nature. [13]
- **Batch Normalization:** The normalization layer used here is Batch Normalization (BN), which is applied after each convolutional operation and before the activation function (ReLU). BN works by normalizing the output of the convolution across the mini-batch, ensuring that the activations have a consistent distribution during training.
- **Regularization:** In ResNet50, regularization is very important since the neural network is so deep that it's very likely to overfit, especially when it works with smaller, unbalanced datasets such as medical images. [14].

C. Fine-Tuning Strategy

To adapt the pre-trained ResNet50 to the dermatological domain (real world conditions) while preserving generic features, we have adopted some selective fine-tuning approach. The first 140 layers of the base model were frozen to retain low-level features learned on ImageNet, while the remaining layers are trained to capture more abstract and task specific features. The entire model was trained in a single phase as 140 layers are frozen and we only need to training remaining layers using the an optimizer with an initial learning rate of 10^{-3} , along with regularisation techniques [15].

D. Model Architecture

We have used ResNet50 model and it is pre-trained on ImageNet. The architecture is designed to use generic visual features while adapting to skin disease domain through selective fine tuning and using a custom classification as per our need. ResNet50 architecture consist of 50 layers oraganised into five main stages: an initial convolutional layer, followed by 4 residual blocks, The early layer capture low level feature

such as edges, corners and colour blobs. The remaining layer are made trainable. These deeper layer are responsible for learning more abstract, domain related features, which allows model to adapt targeted task. After base model, we create a custom classifier head to map the extracted features to the 22 skin disease classes. The head consist of Global Average Pooling which reduces feature map into a single value and the dropout layer randomly deactivate few neurons during training, forcing the network to learn different representation and thus reducing dependence on individual feature.

E. Evaluation

In this qualitative and quantitative evaluation section of the optimized ResNet50 model, we have used standard classification metrics, analyzed the confusion matrix, and examined the learning dynamics to check the effectiveness of optimization [16].

TABLE I
PER-CLASS CLASSIFICATION PERFORMANCE ON THE TEST SET.

Class Name	Precision	Recall	F1-score	Support
Acne	0.28	0.76	0.41	95
Actinic Keratosis	0.59	0.43	0.50	106
Benign tumors	0.26	0.66	0.37	172
Bullous	0.50	0.01	0.03	76
Candidiasis	0.00	0.00	0.00	35
Drug Eruption	0.50	0.58	0.54	99
Eczema	0.31	0.64	0.42	162
Infestations/Bites	0.00	0.00	0.00	92
Lichen	0.40	0.07	0.12	85
Lupus	0.00	0.00	0.00	45
Moles	0.00	0.00	0.00	58
Psoriasis	0.39	0.28	0.33	131
Rosacea	0.00	0.00	0.00	38
Seborrheic keratoses	0.66	0.43	0.52	72
Skin Cancer	0.32	0.19	0.24	117
Sun/Sunlight Damage	0.00	0.00	0.00	53
Tinea	0.17	0.23	0.20	158
Vascular Tumors	0.00	0.00	0.00	86
Vasculitis	0.00	0.00	0.00	280
Total				1960

1) *Evaluation Metrics*: We have evaluated the model using standard classification metrics derived from the confusion matrix. For each class c (with $c = 1, 2, \dots, C$), let TP_c , FP_c , and FN_c denote the number of true positives, false positives, and false negatives, respectively. The per-class precision, recall, and F1-score are defined as:

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c} \quad (1)$$

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c} \quad (2)$$

$$F1_c = 2 \cdot \frac{\text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c} \quad (3)$$

$$\text{Accuracy} = \frac{\sum_{c=1}^C TP_c}{N_{\text{total}}} \quad (4)$$

where N_{total} is the total number of test samples.

To summarise performance across the imbalanced set of 22 classes, we compute both macro-averaged and weighted-averaged metrics. The macro-average F1-score treats each class equally:

$$F1_{\text{macro}} = \frac{1}{C} \sum_{c=1}^C F1_c. \quad (5)$$

F1-score is harmonic mean of Precision and Recall. The weighted-average F1-score weights each class by its support i.e. number of true samples:

$$F1_{\text{weighted}} = \frac{\sum_{c=1}^C TP_c \cdot F1_c}{\sum_{c=1}^C TP_c}. \quad (6)$$

2) *Confusion Matrix*: The confusion matrix provide a detailed view of misclassification. Many samples from minority classes are incorrectly assigned to few high support classes which makes the majority classes dominating in prediction. Some clinically similar conditions are confusing as they reflect the visual overlap of scaly lesions. And some rows shows no true positives, their prediction are spread across other classes. The confusion matrix highlights the problem of the class imbalance.

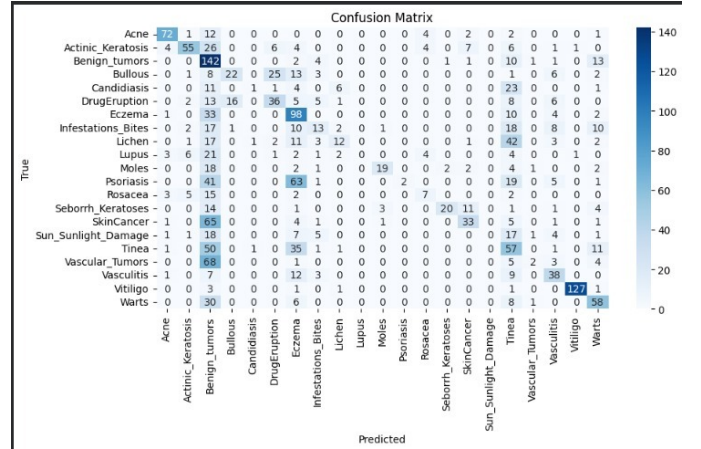


Fig. 3. Confusion Matrix

3) *Loss Curves and Accuracy Graphs*: The training and validation accuracy and loss curves over 50 epochs training period shows that the training accuracy increases steadily while validation accuracy plateaus early, here the widening gap suggest that there is a mild overfitting. The loss in validation stabilises after 10 epochs approximately and the learning rate scheduler shows there is a reduction in learning rate when validation loss stops improving. Early stopping prevent further overfitting by stoping training so that best model is restored.

4) Hardware Configuration:

- **Processor**: Intel Core i7 / AMD Ryzen 7 (Equivalent or higher).

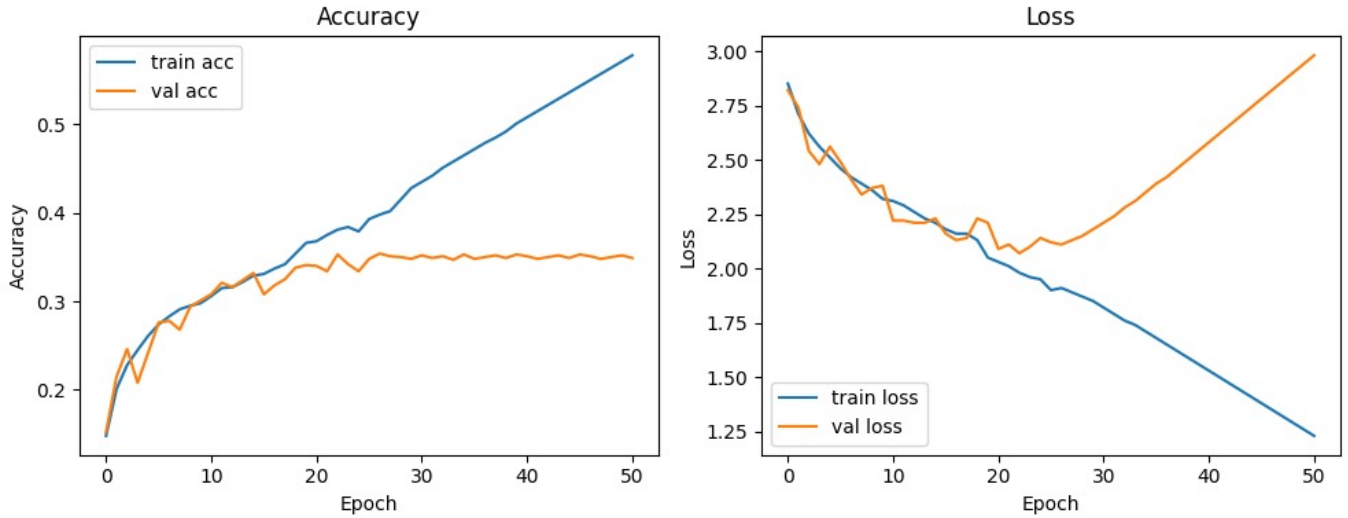


Fig. 4. Accuracy and Loss Graph

- **GPU:** NVIDIA T4 GPU (via Google Colab Pro) for high-speed tensor computations.
- **VRAM:** 16GB High-Bandwidth Memory (HBM).
- **System RAM:** 12.7GB (Standard Colab Environment).

5) Software Environment:

- **Operating System:** Linux-based cloud environment.
- **Programming Language:** Python 3.10.
- **Deep Learning Framework:** TensorFlow 2.x and Keras API.
- **Data Analysis Libraries:** NumPy, Pandas, and Matplotlib.
- **Documentation Tools:** LaTeX code and Overleaf for IEEE formatted conference paper.

IV. RESULTS

In the result section we will cover all the parameter and factors involved in the calculating the result. It represent the quantitative and qualitative evaluation of the optimised ResNet50 model.

A. Overall Performance

The model has achieved a test accuracy of 36.4%. The macro-average precision, recall, and F1-score were each 0.27, while the weighted-average F1-score was 0.31. These values are significantly higher than random guessing, showcasing that the model has learned meaningful patterns. However, the low macro-averages indicate severe difficulty in classifying minority classes.

B. Per-Class Metrics

The table in the methodology section represents precision, recall, and F1-score for each of the 22 classes, along with test set support. Classes with zero recall (e.g., Candidiasis, Infestations/Bites, Lupus, Moles, Rosacea, Sun/Sunlight Damage, Vascular Tumors, Vasculitis) were never correctly identified.

And classes with larger support, such as Benign tumors (recall 0.66) and Eczema (recall 0.64), achieved moderate recall but suffered low precision, indicating that many predictions were false positives.

C. Confusion Matrix Analysis

- **Majority-class bias:** The model tends to predict a small set of high-support classes (e.g., Benign tumors, Eczema, Tinea) for many test samples, causing rows of minority classes to be spread across these columns.
- **Clinically similar confusions:** Tinea (class 16) is frequently misclassified as Eczema (class 6) and Seborrheic keratoses (class 13), reflecting visual overlap of scaly lesions. Actinic keratosis (class 1) and Sun/Sunlight damage (class 15) also show mutual confusion.
- **Zero-performance rows:** Several classes have no true positive predictions, meaning the model fails entirely to recognise those conditions which has lesser size of the dataset.

D. Training Dynamics

The accuracy-loss graph shows that the training and validation accuracy/loss over 50 epochs. Training accuracy steadily increases, while validation accuracy plateaus after approximately 10 epochs. The widening gap indicates mild overfitting, which is lowered by early stopping and learning rate reduction. The validation loss is stabilized, and the best model is restored from the epoch with the lowest validation loss.

V. CONCLUSION

In this paper, we present an optimised ResNet50 model for multi-class skin disease classification. The optimisation strategy combined transfer learning technique with selective fine-tuning, data augmentation, regularisation classifier head with dropout, early stopping, and learning rate scheduling. [17]

The model was evaluated on a real-world dataset of 22 skin disease categories exhibiting severe class imbalance.

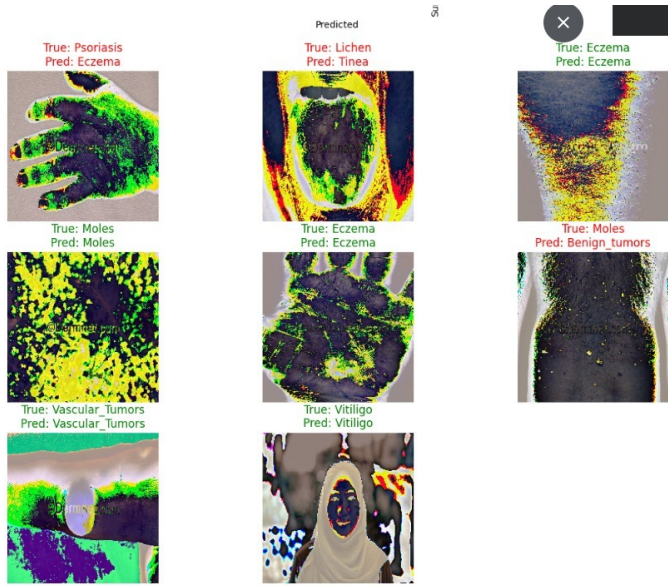


Fig. 5. Output Images

Our results shows that the optimised model which achieves an overall accuracy of 36.4%, with a macro-average F1-score of 0.27. While the model performs reasonably on a few majority classes, it fails entirely on many minority conditions. The confusion matrix reveals that misclassifications are accumulated among similar diseases and that the model's predictions are biased toward high-support classes. These findings highlights the critical challenge of class imbalance in medical image classification and demonstrate that even with advanced optimisation techniques, standard fine-tuning is insufficient when rare diseases have less representation. The main contributions of this work are:

- A systematic fine-tuning and regularisation pipeline for ResNet50 tailored to skin disease classification.
- A detailed analysis of per-class performance and confusion patterns, providing insights for future research.

VI. LIMITATIONS

There are some limitation in this project specifically in the training the model and dataset used which leads to have some huge impact in the result section like: The dataset used here was not perfectly balanced and no explicit class balancing mechanism was used. This leads to poor performance in case of minority classes. The model was trained on clinical images with varying conditon so generalisation to other dataset may be limited.

VII. FUTURE WORK

Future work will focus on improving the performance of the proposed model and making it easy to use and also available to all of us. But the major work should be done particularly for

minority classes and making this model, through the following directions:

- Manage cost-sensitive learning or focus on increasing dataset of rare diseases for better training and accuracy.
- Explore other models so that we can optimise current one by extracting important features from other models and improving the performance.
- Evaluating the model on external datasets to assess generalisability. We can also find defect or mistake in our current model by working on external datasets.
- Integrate explainability methods such as Grad-CAM to provide justification for the predictions, which is essential for clinical acceptance.
- Deploy the model as a mobile-based screening tool for preliminary diagnosis in resource-constrained environment.

REFERENCES

- [1] R. Yadav and A. Bhat, "A systematic literature survey on skin disease detection and classification using machine learning and deep learning," *Multimedia Tools and Applications*, vol. 83, no. 32, pp. 78 093–78 124, 2024.
- [2] S. Inthiyaz, B. R. Altahan, S. H. Ahammad, V. Rajesh, R. R. Kalangi, L. K. Smirani, M. A. Hossain, and A. N. Z. Rashed, "Skin disease detection using deep learning," *Advances in Engineering Software*, vol. 175, p. 103361, 2023.
- [3] V. Anand, S. Gupta, D. Koundal, S. R. Nayak, J. Nayak, and S. Vimal, "Multi-class skin disease classification using transfer learning model," *International Journal on Artificial Intelligence Tools*, vol. 31, no. 02, p. 2250029, 2022.
- [4] O. N. Pise and S. Borde, "Skin disease detection using machine learning," *Int. J. Eng. Res. Technol.(IJERT)*, vol. 9, pp. 64–68, 2020.
- [5] O. Jaiyeoba, O. Jaiyeoba, E. Ogbuju, and F. Oladipo, "Ai-based detection techniques for skin diseases: A review of recent methods, datasets, metrics, and challenges," *Journal of Future Artificial Intelligence and Technologies*, vol. 1, no. 3, pp. 318–336, 2024.
- [6] M. Sazzadul Islam Prottasha, S. Mahjabin Farin, M. Bulbul Ahmed, M. Zihadur Rahman, A. Kabir Hossain, and M. Shamim Kaiser, "Deep learning-based skin disease detection using convolutional neural networks (cnn)," in *The Fourth Industrial Revolution and Beyond: Select Proceedings of IC4IR+*. Springer, 2023, pp. 551–564.
- [7] P. Kumar, S. Arora, and S. Arora, "Real-time sugarcane bud detection using efficient lightweight convolutional blocks," *Arabian Journal for Science and Engineering*, 2025. [Online]. Available: <https://doi.org/10.1007/s13369-025-10906-3>
- [8] —, "Real time sugarcane stem health detection with yolov8 tinier and repncsp elan," *Scientific Reports*, vol. 15, no. 1, p. 43672, 2025.
- [9] R. Alharith, A. O. Ibrahim, N. Wahid, R. Ghazali, and A. Elsafi, "Comparative analysis of resnet models for skin cancer diagnosis: performance evaluation and insights," in *International Conference on Soft Computing and Data Mining*. Springer, 2024, pp. 370–378.
- [10] M. Sharma, B. Jain, C. Kargeti, V. Gupta, and D. Gupta, "Detection and diagnosis of skin diseases using residual neural networks (resnet)," *International Journal of Image and Graphics*, vol. 21, no. 05, p. 2140002, 2021.
- [11] M. Elshahawy, A. Elnemr, M. Oproescu, A.-G. Schiopu, A. Elgarayhi, M. M. Elmogy, and M. Sallah, "Early melanoma detection based on a hybrid yolov5 and resnet technique," *Diagnostics*, vol. 13, no. 17, p. 2804, 2023.
- [12] S. P. Choy, B. J. Kim, A. Paolino, W. R. Tan, S. M. L. Lim, J. Seo, S. P. Tan, L. Francis, T. Tsakok, M. Simpson *et al.*, "Systematic review of deep learning image analyses for the diagnosis and monitoring of skin disease," *NPJ Digital Medicine*, vol. 6, no. 1, p. 180, 2023.
- [13] P. V. Raghavendra, C. Charitha, K. G. Begum, and V. S. Prasath, "Deep learning-based skin lesion multi-class classification with global average pooling improvement," *Journal of Digital Imaging*, vol. 36, no. 5, pp. 2227–2248, 2023.

- [14] S. Benyahia, B. Meftah, and O. L  zoray, "Multi-features extraction based on deep learning for skin lesion classification," *Tissue and Cell*, vol. 74, p. 101701, 2022.
- [15] P. Agarwal and P. Kumar, "Automated bone fracture detection using yolov8," in *2025 3rd International Conference on Disruptive Technologies (ICDT)*, 2025, pp. 1256–1260.
- [16] K. Pfisterer, L. E. Shaw, D. Symmank, and W. Weninger, "The extracellular matrix in skin inflammation and infection," *Frontiers in cell and developmental biology*, vol. 9, p. 682414, 2021.
- [17] M. Alruwaili and M. Mohamed, "An integrated deep learning model with efficientnet and resnet for accurate multi-class skin disease classification," *Diagnostics*, vol. 15, no. 5, p. 551, 2025.